电子科技大学
格拉斯哥学院
Glasgow College, UESTC

UESTC 3010: Team Design Project and Skills

Lecture 1a

Course Information

Course Aims

- The aim of this course is:
  - Build the experience of **working in a team** to design and construct an integrated electronic system that must perform a specific function within a given budget.

Intended Learning Outcomes

1. Analyse technical requirements to develop and justify an overall design plan within a given budget;
2. Design, construct and validate electronic hardware to perform specific functions;
3. Apply a project plan using project management tool (such as PERT), defining milestones and measuring achievement against such milestones, and evaluate a risk mitigation plan;
4. Demonstrate teamwork skills by working effectively within a team;
5. Produce a technical report that outline the analysis and present the results clearly.

Teaching and Supervision

- Teaching and supervision team consists of seven members of staff:

Dr. Abdullah Al-Khalidi (coordinator)

Dr. Amir Parnianifard

Prof. João Ponciano

Teaching and Supervision

- Teaching and supervision team consists of six members of staff:

Dr. Hanaa Abumarshoud

Dr. Wasim Ahmad

Dr. Hasan Abbas

Timetable

- There will be three lectures this semester.
- All lecture slides will be available on Moodle.
- The lecture timetable is as follows:

Sessions	Monday	Tuesday	Wednesday	Thursday	Friday
	2026/03/02	2026/03/03	2026/03/04	2026/03/05	2026/03/06
Session 1-2 08:30–10:05					
Session 3-4 10:20–11:55					
Session 5-6 14:30–16:05		Lecture ◊ Liren Building B218 ◊ IE & CE1 ◊		Lecture ◊ Liren Building B218 ◊ IE & CE1 ◊	
Session 7-8 16:20–17:55	Lecture ◊ Liren Building B218 ◊ ME&CE2,CE3,CE4 ◊	Lecture ◊ Liren Building B218 ◊ ME&CE2,CE3,CE4 ◊	Lecture ◊ Liren Building B218 ◊ IE & CE1 ◊	Lecture ◊ Liren Building B218 ◊ ME&CE2,CE3,CE4 ◊	
Session 9-10 19:30–21:05					

Reference Books

Authors	Book title	Publishers	ISBN
P Horowitz and W Hill	The Art of Electronics	Cambridge	2nd Edition, ISBN 521370957
A S Sedra and K C Smith	Microelectronic Circuits	Oxford	5th Edition, ISBN 195142527
B Baker	A Baker's Dozen: Real Analog Solutions for Digital Designers	Newnes	ISBN 750678194
W Jung	Op Amp Applications Handbook	Newnes	ISBN 750678445
W Kester	Mixed-signal and DSP Design Techniques	Newnes	ISBN 750676116
W Kester	Data Conversion Handbook	Newnes	ISBN 750678410
R Mancini	Op Amps for Everyone	Texas Instruments	http://www.ti.com/lit/an /slod006b/slod006b.pdf
B Carter	Op Amps for Everyone	Newnes	ISBN-10: 0123914957, ISBN-13: 978-0123914958
R Toulson, T Wilmshurst	Fast and Effective Embedded Systems Design: Applying the ARM mbed	Newnes	ISBN-10: 0080977685, ISBN-13: 978-0080977683
R M Woefle	A new guide for better technical presentations	IEEE	ISBN 879422831
D F Beer	Writing and speaking in the technology professions	IEEE	ISBN 087942284X
B Dobkin, J Williams	Analog Circuit Design: A Tutorial Guide to Applications and Solutions	Newnes	ISBN 9780123851857
J H Davies	MSP430 Microcontroller Basics	Newnes	ISBN 9780750682763
J E Salt, R Rothery	Design for Electrical and Computer Engineers	Wiley	978-0471391463

Course Material

- The course is supported by a Moodle site.
- All the relevant material will be available on the course Moodle page and can be accessed at:

https://moodle.gla.ac.uk/course/view.php?id=53118

Course Assessment

%	Assessment Type	When	Details
10	Initial design report (5%) and laboratory notebook (5%)	Week 8	Initial design, responsibilities, lab notebook, progress discussion.
25	Final demonstration	Week 15	Demonstration of a drone that performs certain tasks on a given patio.
40	Final report presentation	Week 15	50-minute oral team presentation (25 minutes PPT + 25 minutes Q&A).
25	Final Team report	Week 16	Final team report.

Requirements for Grade

- To receive a grade at the end of the course, you must:
  - attend all the assessments – initial design meeting, final group demonstration and presentation
  - submit final group and individual reports before the due date
- If you fail to fulfil any of the above requirement, you will be given "CW".
- CW (Credit Withheld) means that you have not completed some part of the assessment (exam, laboratory report, etc) but can still do so before the next academic year. Contact the course lecturer if you are in doubt as to what you need to do.

Group formation

- You are required to work as a team to design and complete the given tasks
- 5-students in each team:
  - You are expected to select your team members and form a team.
  - Each team **should** have at least **one** member from each of the **three degree programmes** (IE, CE & ME).
  - You need to submit your team form on Moodle by **Monday 9th March 2026.**

Group Design Project

- Each team will be required to submit:
  - An initial design report with a short description and a lab notebook.
  - A demo (group).
  - A presentation (individual & group).
  - Final report (group) and final lab notebook.

Team Organisation

- Teamwork
- Team leader and other team members role and responsibilities
- Tasks should be identified/categorized
- Task(s) assignment to individuals or subgroups
- Planning and time management
- Think outside the box
- Have a plan A, B, and may be C

Microsoft Teams

- Microsoft Teams will be the main mean of communication and interaction on the TDPS course.
- Once the teams are finalised, students will be added to Microsoft Teams and each team will have a dedicated private channel.
- Teams will be divided under **six main** groups, teams under Group-1 will be supervised by Dr. Abdullah Al-Khalidi, teams under Group-2 will be supervised by Dr. Amir Parnianifard, teams under Group-3 will be supervised by Wasim Ahmad, teams under Group-4 will be supervised by Dr. Hanaa Abumarshoud, teams under Group-5 will be supervised by Prof. João Ponciano, and Group-6 will be supervised by Dr. Hasan Abbas.
- The General Channel will be for General announcements only made by teaching staff. If you have any questions, please ask them using your private Teams channel.

Key dates

Description	Date
Submission of the team form on Moodle	Monday 9 March 2026
Submission of initial report and notebook	Sunday 19 April 2026
Mid-term progress assessment	Monday - Friday 20-24 April 2026
Mock demonstration	Monday – Friday 25–29 May 2026
Final demonstration	Saturday 6 June 2026
Presentation slides submission	Sunday 7 June 2026
Final team presentations	Monday - Friday 8 - 12 June 2026
Final report submission	Sunday 21 June 2026